

Bohr Model

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Forces

Bohr identified the two forces that act on the electron in the atom. The centripetal force is the attractive force that pulls the electron toward the nucleus. The centrifugal force is the force that pushes the electron away from the nucleus. The forces are balanced. Otherwise, the electron would be ejected from the atom or would crash into the nucleus.

$$F_{centrifugal} = \frac{mv^2}{r}$$

$$F_{centripetal} = \frac{Ze^2}{kr^2}$$

$e = 1.602 \times 10^{-19}C$ is the fundamental charge of an electron or proton.

k = constant (may also be written in the numerator)

$$\frac{mv^2}{r} = \frac{Ze^2}{kr^2}$$

Quantization

To account for the quantization, Bohr restricted the angular momentum according to "n" or the energy levels.

$$mvr = \frac{nh}{2\pi}$$

Bohr Radius

Solve for v using the angular momentum relationship.

$$v = \frac{nh}{2mr\pi}$$

square both sides

$$v^2 = \frac{n^2h^2}{4m^2r^2\pi^2}$$

finally substitute for v^2 in the centrifugal force equation

$$\frac{mn^2h^2}{4m^2r^3\pi^2} = \frac{Ze^2}{kr^2}$$

make cancellations to simplify the expressions

$$\frac{n^2h^2}{4mr\pi^2} = \frac{Ze^2}{k}$$

$$\text{Solve for } r: r = \frac{h^2kn^2}{4me^2\pi^2Z}$$

The constants combine to give the Bohr radius, $a_o = 52.9$ and $r = a_o \frac{n^2}{Z}$.

Energy

Recall that the total energy is a sum of the kinetic and potential energy. $E_{total} = \frac{1}{2}mv^2 + PE$

The potential energy $= -\frac{Ze^2}{kr}$ where k is a constant.

Using the relationship between the forces, we can solve for v^2 .

$$\frac{mv^2}{r} = \frac{Ze^2}{kr^2}$$

$$v^2 = \frac{Ze^2}{krm}$$

Making the substitution into the equations for v^2 and potential energy:

$$E_{total} = \frac{1}{2}m \frac{Ze^2}{krm} - \frac{Ze^2}{kr}$$

The m term cancels in the kinetic energy component

$$E_{total} = \frac{1}{2} \frac{Ze^2}{kr} - \frac{Ze^2}{kr}$$

Taking the difference gives:

$$E_{total} = -\frac{1}{2} \frac{Ze^2}{kr}$$

Finally substitute for r, using the expression $r = a_o \frac{n^2}{Z}$.

$E_{total} = -\frac{1}{2} \frac{a_o Z^2 e^2}{kn^2}$ and the constants combine to simplify the expression to give:

$$E_{total} = -2.18 \times 10^{-18} J \frac{Z^2}{n^2}.$$